(Please write your Enrollment Number) Enrollment No. 19401172023

MID-TERM EXAMINATION

Course Name: B. Tech, Semester: 5th

October, 2025 OFF LINE mode

Subject Code: BCS 303	Subject: Theory of Computation
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1	Q1 is compulsory.	(2.5*4=10)	CO Mapping
	a) Define the following terms with examples: Language, Grammar, and Deterministic Finite Automata.		CO1
	b) Explain the equivalence of DFA and NDFA with a simple example.		CO1
	c) Define Context Free Grammar (CFG) and give an example of a production rule.		CO2
	d) What is a Pushdown Automaton (PDA)? Why only stack is used as a memory element in PDA and why not any other Data Structure?		CO2
	UNIT I		CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Construct a DFA that accepts all strings over the alphabet {0,1} that end with "01". Draw the state diagram and explain the transitions.		CO1
	b) What is the Pumping Lemma for regular languages? State the lemma and explain its significance with an example.		CO1
	c) Construct a NFA for (0*1*) and covert this NFA into equivalent DFA		CO1
	UNIT II		CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) Construct a grammar for $\{ww^R\}$. What is Left Derivation and write derivation		CO2
	b) Consider the two languages $\{0^n1^n$ where n>=0}and (0*1*). Which of them is a regular language and which is not a regular language. Prove by giving a proper proof.		CO2
	c) Draw the parse tree for the string "aaabb" generated by the grammar: S→aSb|ϵ Also explain ambiguity in context free grammars. What is inherently ambiguous grammar?		CO2

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